

WADF104 - Cybersecurity Foundations
Lab 1: Network and Service Reconnaissance
with Nmap and WhatWeb

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Platform	Kali Linux + Metasploitable2

1. Scenario and Authorisation

This report documents authorised reconnaissance performed against a Metasploitable2 virtual machine within an isolated VirtualBox host-only lab network, as a junior security analyst exercise. The objective was to identify the target, enumerate exposed network services, determine service versions, examine the likely operating system, and fingerprint the exposed web service, in preparation for the Lab 2 web application testing exercise.

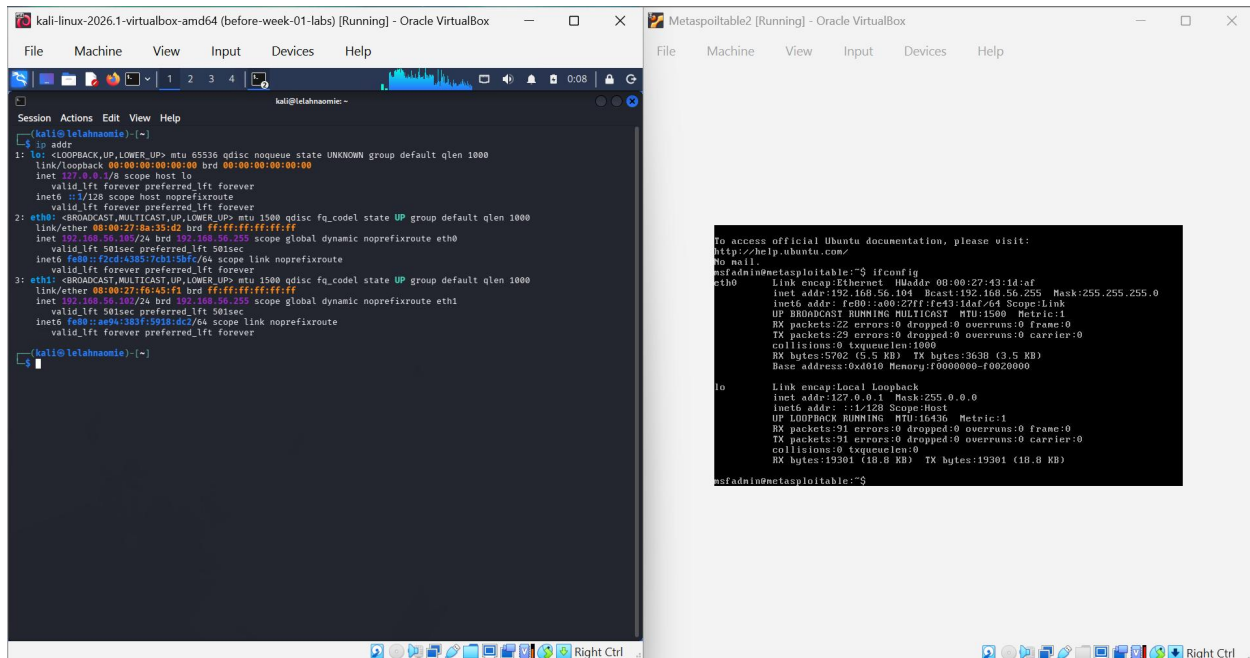
All scanning in this report was performed exclusively against the assigned lab target (192.168.56.104) from the Kali attacker VM (192.168.56.105), both isolated on a VirtualBox Host-only network with no external connectivity.

2. Evidence and Reconnaissance Findings

2.1 Target Identification and Connectivity

Both VMs were placed on a shared VirtualBox Host-only Adapter network after an initial NAT misconfiguration (both machines sharing the same 10.0.2.15 address) was corrected. Final addressing:

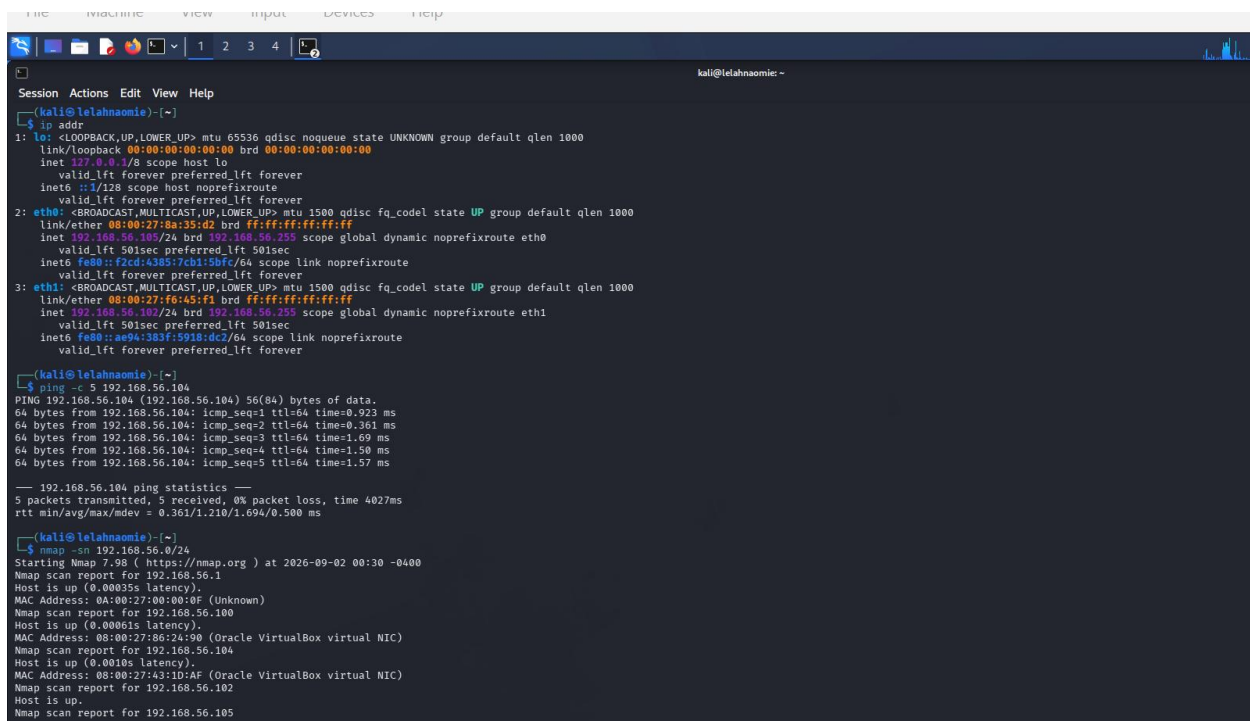
Kali (attacker):192.168.56.105
Metasploitable2 (target): 192.168.56.104



Connectivity was confirmed with:

ping -c 4 192.168.56.104

Result: reachable, 0% packet loss (see note below).

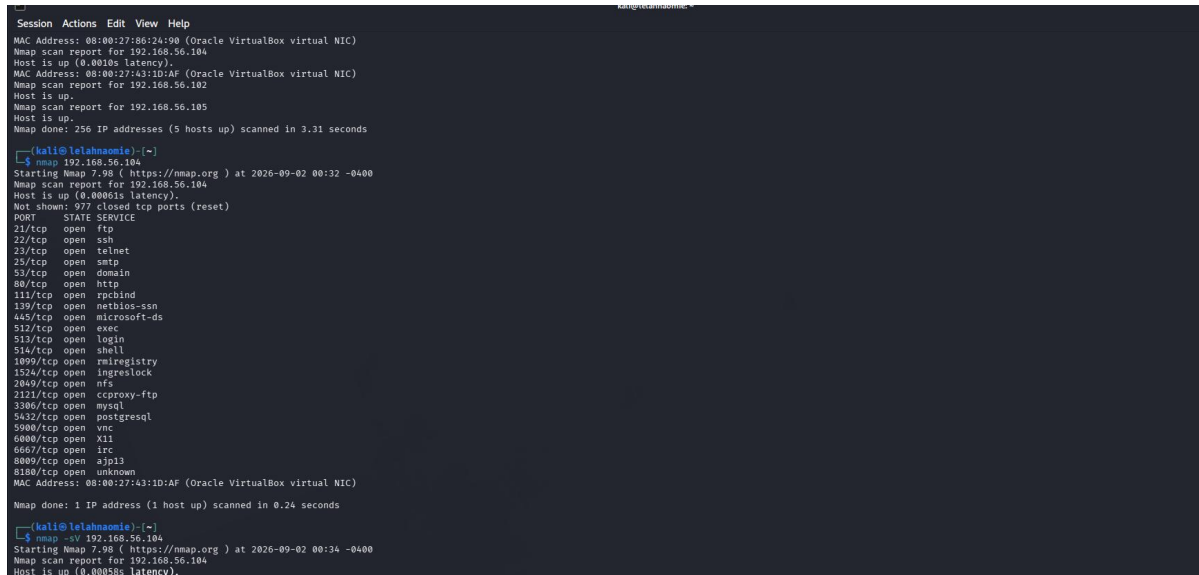


2.2 Host Discovery

A ping-sweep of the /24 subnet identified 5 live hosts, including the confirmed target 192.168.56.104. A default Nmap scan against the target then listed its top-1000-port open services.

```
nmap -sn 192.168.56.0/24
```

```
nmap 192.168.56.104
```



```
Session Actions Edit View Help
MAC Address: 08:00:27:86:24:90 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.104
Host is up (0.0002s latency).
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102
Host is up.
Nmap scan report for 192.168.56.105
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 3.31 seconds

(kali@lelahnaomie)~$
(kali@lelahnaomie)~$ nmap 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 00:32 -0400
Nmap scan report for 192.168.56.104
Host is up (0.00061s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  cproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8080/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds

(kali@lelahnaomie)~$
(kali@lelahnaomie)~$ nmap -sV 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 00:34 -0400
Nmap scan report for 192.168.56.104
Host is up (0.00055s latency).
```

Figure 1. Host discovery sweep (5 hosts up) followed by the default Nmap scan of the target.

2.3 Service and Version Detection (-sV)

Service/version detection identified specific application versions behind each open port, e.g. vsftpd 2.3.4, OpenSSH 4.7p1, Postfix smtpd, ISC BIND 9.4.2, and Apache httpd 2.2.8 (Ubuntu) DAV/2.

```
nmap -sV 192.168.56.104
```

```
nmap -sV --version-intensity 9 192.168.56.104
```

```
kali@lelahnaomie: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds  
  
(kali@lelahnaomie)-[~]  
$ nmap -sV 192.168.56.104  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 00:34 -0400  
Nmap scan report for 192.168.56.104  
Host is up (0.00058s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rshcd  
513/tcp   open  login        OpenBSD or Solaris rlogind  
514/tcp   open  shell        Netkit rshd  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 12.12 seconds  
  
(kali@lelahnaomie)-[~]  
$ nmap -sV --version-intensity 9 192.168.56.104  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 00:37 -0400  
Nmap scan report for 192.168.56.104  
Host is up (0.00062s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd
```

Figure 2. Full `-sV` service/version table, followed by the maximum-intensity re-scan for comparison.

2.4 OS Detection (-O) and Aggressive Scan (-A)

OS fingerprinting identified the target as running Linux kernel 2.6.9-2.6.33 (general purpose device), CPE `cpe:/o:linux:linux_kernel:2.6`. The aggressive scan combined OS detection, version detection, default scripts and traceroute, and additionally revealed that anonymous FTP login is allowed and that the FTP/SSH/SMTP services expose detailed banner and certificate information.

```
sudo nmap -O 192.168.56.104
```

```
sudo nmap -A 192.168.56.104
```

```
File Machine View Input Devices Help

kali@telahnaomie: ~
Session Actions Edit View Help
23/tcp open telnet Linux telnetd
25/tcp open smtp Postfix smtpd
53/tcp open domain ISC BIND 9.4.2
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp open rpcbind 2 (RPC #100000)
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp open exec netkit-rsh rexecd
513/tcp open login
514/tcp open shell Netkit rshd
1099/tcp open java-rmi GNU Classpath gmrregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-Subuntu5
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open vnc VNC (protocol 3.3)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.95 seconds

kali@telahnaomie: ~
$ sudo nmap -O 192.168.56.104
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 00:41 -0400
Nmap scan report for 192.168.56.104
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
```

Figure 3. OS detection result (Linux 2.6.9-2.6.33) followed by the start of the -A aggressive scan.

```
kali@telahnaomie: ~
Session Actions Edit View Help
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open cccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.68 seconds

kali@telahnaomie: ~
$ sudo nmap -A 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 00:43 -0400
Nmap scan report for 192.168.56.104
Host is up (0.0018s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.56.105
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp_commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BI
```

Figure 4. -A aggressive scan detail: anonymous FTP login allowed, SSH host keys, Postfix SMTP commands and SSLv2 cipher enumeration.

```

Session Actions Edit View Help
| thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
| Not valid after: 2010-04-16T14:07:45
5900/tcp open  vnc          VNC (protocol 3.3)
| vnc-info:
|   Protocol version: 3.3
|   Security types:
|   VNC Authentication (2)
6000/tcp open  x11            (access denied)
6667/tcp open  irc            UnrealIRCd
8009/tcp open  ajp13          Apache Jserv (Protocol v1.3)
|_ ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http           Apache Tomcat/Coyote JSP engine 1.1
|_ http-favicon: Apache Tomcat
|_ http-server-header: Apache-Coyote/1.1
|_ http-title: Apache Tomcat/5.5
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-09-02T00:43:30-04:00
|_  clock-skew: mean: 59m58s, deviation: 2h00m00s, median: -1s
|_  smb2-time: Protocol negotiation failed (SMB2)
|_  nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_  smb-security-mode:
|    account_used: <blank>
|    authentication_level: user
|    challenge_response: supported
|_  message_signing: disabled (dangerous, but default)

TRACEROUTE
HOP RTT ADDRESS
1 1.01 ms 192.168.56.104

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.13 seconds

```

Figure 5. -A aggressive scan (continued): VNC info, Tomcat/AJP detail, OS summary, traceroute, and SMB security mode showing message signing disabled ("dangerous, but default").

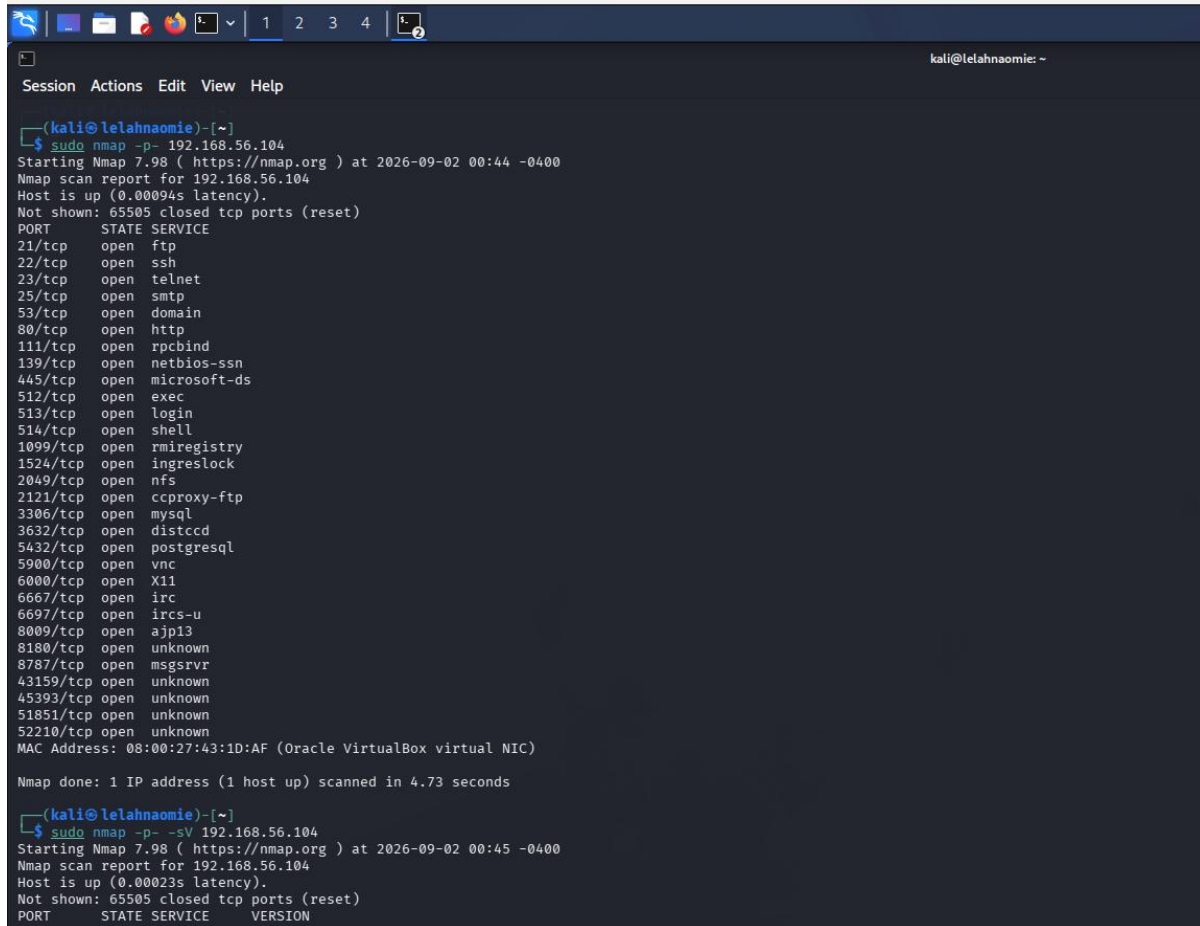
2.5 Full TCP Port Scan (-p-)

Scanning all 65535 TCP ports (rather than the default top 1000) revealed 28 open ports in total, including several high-numbered ports not visible in the default scan: 43159 (java-rmi), 45393 (mountd), 51851 (nlockmgr), and 52210 (status).

```

sudo nmap -p- 192.168.56.104
sudo nmap -p- -sV 192.168.56.104
time sudo nmap -p- -T4 192.168.56.104

```



```
(kali@lelahnaomie)-[~]
$ sudo nmap -p- 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 00:44 -0400
Nmap scan report for 192.168.56.104
Host is up (0.00094s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
43159/tcp open  unknown
45393/tcp open  unknown
51851/tcp open  unknown
52210/tcp open  unknown
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.73 seconds

(kali@lelahnaomie)-[~]
$ sudo nmap -p- -sV 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 00:45 -0400
Nmap scan report for 192.168.56.104
Host is up (0.00023s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          0.9.5
22/tcp    open  ssh          OpenSSH_8.2p1 Ubuntu-0ubuntu0.2, Ubuntu 8.2p1
23/tcp    open  telnet       0.52
25/tcp    open  smtp         postfix
53/tcp    open  domain       BIND 9.18.24
80/tcp    open  http         Apache/2.4.18 (Ubuntu)
111/tcp   open  rpcbind      2.4.2
139/tcp   open  netbios-ssn  Samba 4.15.13
445/tcp   open  microsoft-ds
512/tcp   open  exec         2.0.1
513/tcp   open  login        2.1.1
514/tcp   open  shell        2.0.1
1099/tcp  open  rmiregistry  1.2.3
1524/tcp  open  ingreslock   1.1.1
2049/tcp  open  nfs          4.2.0
2121/tcp  open  ccproxy-ftp  3.11.1
3306/tcp  open  mysql        8.0.33-0ubuntu0.22.04.1
3632/tcp  open  distccd      3.10
5432/tcp  open  postgresql   15.3-0ubuntu0.22.04.1
5900/tcp  open  vnc          5.10
6000/tcp  open  X11          7.7
6667/tcp  open  irc          Libera 0.0.19
6697/tcp  open  ircs-u       Libera 0.0.19
8009/tcp  open  ajp13        4.2.13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr      5.0.0
43159/tcp open  unknown
45393/tcp open  unknown
51851/tcp open  unknown
52210/tcp open  unknown
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.73 seconds
```

Figure 6. Full 65535-port TCP scan showing all 28 open ports, including the high-numbered ports missed by the default scan.


```

kali@lelahnaomie: ~
Session Actions Edit View Help
445/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp open exec netkit-rsh rshd
513/tcp open login OpenBSD or Solaris rlogind
514/tcp open shell Netkit rshd
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
3632/tcp open distccd distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open vnc VNC (protocol 3.3)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
6697/tcp open irc UnrealIRCd
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1
8787/tcp open drb Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbb)
43159/tcp open java-rmi GNU Classpath grmiregistry
45393/tcp open mountd 1-3 (RPC #100005)
51851/tcp open nlockmgr 1-4 (RPC #100021)
52210/tcp open status 1 (RPC #100024)
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.IAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 131.05 seconds

(kali@lelahnaomie)~]
$ time sudo nmap -p- -T4 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 00:48 -0400
Nmap scan report for 192.168.56.104
Host is up (0.00033s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry

```

Figure 7. Full-port scan with version detection (-p- -sV) and the -T4 timing-template re-scan for speed comparison.

2.6 Selected Ports and Port Range Scans

Focused scans against known service ports and the 1-1024 well-known range were used to confirm findings quickly without repeating a full scan.

```
nmap -p 21,22,23,25,80,139,445 192.168.56.104
nmap -p 1-1024 192.168.56.104
```

2.7 UDP Scanning

A full default UDP scan (-sU) took 1017 seconds due to the lack of a UDP handshake, forcing Nmap to wait out timeouts on non-responsive ports. A faster top-20-ports scan with version detection completed in seconds and confirmed ISC BIND 9.4.2 on UDP/53.

```
sudo nmap -sU 192.168.56.104
sudo nmap -sU --top-ports 20 -sV 192.168.56.104
```



```
Session Actions Edit View Help
(kali@lelahnaomie)-[~]
$ nmap -p 1-1024 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 00:48 -0400
Nmap scan report for 192.168.56.104
Host is up (0.00022s latency).
Not shown: 1012 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.27 seconds

(kali@lelahnaomie)-[~]
$ sudo nmap -sU 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 00:48 -0400
Nmap scan report for 192.168.56.104
Host is up (0.0013s latency).
Not shown: 949 closed udp ports (port-unreach), 47 open|filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
111/udp   open  rpcbind
137/udp   open  netbios-ns
2049/udp  open  nfs
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1017.79 seconds

(kali@lelahnaomie)-[~]
$ sudo nmap -sU --top-ports 20 -sV 192.168.56.104
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 01:13 -0400
Nmap scan report for 192.168.56.104
Host is up (0.0011s latency).

PORT      STATE SERVICE VERSION
53/udp    open  domain  ISC BIND 9.4.2
67/udp    closed dhcpcd
68/udp    open|filtered dhcpcd
```

Figure 8. Full UDP scan (1017.79 seconds) and the faster top-20-ports UDP scan with version detection.

2.8 Safe NSE Enumeration

Default NSE scripts (-sC -sV) and targeted scripts (http-title, http-headers, smb-protocols, ssh-hostkey, banner) added detail beyond plain version detection: SMBv1 is still enabled and flagged as dangerous by Nmap's own script, and MySQL/RPC service metadata was fully enumerated.

```
nmap -sC -sV 192.168.56.104
nmap -p 80 --script http-title 192.168.56.104
nmap -p 139,445 --script smb-protocols 192.168.56.104
nmap -p 80 --script http-headers 192.168.56.104
nmap -p 22 --script ssh-hostkey 192.168.56.104
nmap -p 21 -sV --script banner 192.168.56.104
```

```

kali@lelahnaomie: ~
Session Actions Edit View Help
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such
| thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
| Not valid after: 2010-04-16T14:07:45
| sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
| 53/tcp open domain ISC BIND 9.4.2
| dns-nsid:
|   bind.version: 9.4.2
| 80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_ http-title: Metasploitable2 - Linux
| 111/tcp open rpcbind 2 (RPC #100000)
| rpcinfo:
|   program version port/proto service
|   100000 2 111/tcp rpcbind
|   100000 2 111/udp rpcbind
|   100003 2,3,4 2049/tcp nfs
|   100003 2,3,4 2049/udp nfs
|   100005 1,2,3 35354/udp mountd
|   100005 1,2,3 45393/tcp mountd
|   100021 1,3,4 44313/udp nlockmgr
|   100021 1,3,4 51851/tcp nlockmgr
|   100024 1 52210/tcp status
|   100024 1 54374/udp status
| 139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
| 445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
| 512/tcp open exec netkit-rsh rexecd
| 513/tcp open login
| 514/tcp open shell Netkit rshd
| 1099/tcp open java-rmi GNU Classpath grmiregistry
| 1524/tcp open bindshell Metasploitable root shell
| 2049/tcp open nfs 2-4 (RPC #100003)
| 2121/tcp open ftp ProFTPD 1.3.1
| 3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 10
|   Capabilities flags: 43564
|   Some Capabilities: LongColumnFlag, ConnectWithDatabase, SwitchToSSLAfterHandshake, Speaks41ProtocolNew, SupportsCo
| mpression, SupportsTransactions, Support41Auth
```

Figure 9. Default NSE scripts (-sC -sV): SSL/TLS cipher enumeration, DNS version, RPC service table, and MySQL protocol info.

```
Session Actions Edit View Help
Nmap done: 1 IP address (1 host up) scanned in 21.76 seconds

(kali@lelahnaomie)-[~]
$ nmap -p 80 --script http-title 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 01:20 -0400
Nmap scan report for 192.168.56.104
Host is up (0.00055s latency).

PORT      STATE SERVICE
80/tcp    open  http
|_ http-title: Metasploitable2 - Linux
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.48 seconds

(kali@lelahnaomie)-[~]
$ nmap -p 139,445 --script smb-protocols 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 01:20 -0400
Nmap scan report for 192.168.56.104
Host is up (0.00041s latency).

PORT      STATE SERVICE
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:43:1D:AF (Oracle VirtualBox virtual NIC)

Host script results:
| smb-protocols:
|   dialects:
|_    NT LM 0.12 (SMBv1) [dangerous, but default]

Nmap done: 1 IP address (1 host up) scanned in 0.42 seconds

(kali@lelahnaomie)-[~]
$ nmap -p 80 --script http-headers 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 01:21 -0400
Nmap scan report for 192.168.56.104
Host is up (0.00044s latency).

PORT      STATE SERVICE
80/tcp    open  http
|_ http-headers:
|   Date: Wed, 02 Sep 2026 05:21:39 GMT
|   Server: Apache/2.2.8 (Ubuntu) DAV/2
|   X-Powered-By: PHP/5.2.4-2ubuntu5.10
|   Connection: close
|   Content-Type: text/html
|
|_ (Request type: HEAD)
```

Figure 10. Targeted NSE scripts: *http-title*, *smb-protocols* (SMBv1 flagged "dangerous, but default"), and *http-headers*.

2.9 WhatWeb Web Fingerprinting

WhatWeb confirmed the web technology stack from the application layer: Apache 2.2.8 (Ubuntu) with DAV/2, PHP 5.2.4-2ubuntu5.10, and WebDAV version 2. Comparing aggression levels showed that level 4 (maximum aggression) additionally flagged a possible Matomo installation that was not detected at levels 1 or 3 — an example of higher aggression surfacing more (and occasionally speculative) results.

```
whatweb http://192.168.56.104
whatweb -v http://192.168.56.104
whatweb -a 1 http://192.168.56.104
whatweb -a 3 -v http://192.168.56.104
whatweb -a 4 -v http://192.168.56.104
whatweb --follow-redirect=always http://192.168.56.104
whatweb -v http://192.168.56.104 > whatweb-results.txt
```

```
Session Actions Edit View Help
$ whatweb http://192.168.56.104
http://192.168.56.104 [200 OK] Apache[2.2.8], Country[RESERVED][22], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.104], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2u
buntu5.10]

[kali@lelahannia:~]$
$ whatweb -v http://192.168.56.104
WhatWeb report for http://192.168.56.104
Status : 200 OK
Title : Metasploitable2 - Linux
IP : 192.168.56.104
Country : RESERVED, 22
Summary : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
The Apache HTTP Server Project is an effort to develop and
maintain an open-source HTTP server for modern operating
systems including UNIX and Windows NT. The goal of this
project is to provide a secure, efficient and extensible
server that provides HTTP services in sync with the current
HTTP standards.

Version : 2.2.8 (from HTTP Server Header)
Google Dorks: (3)
Website : http://httpd.apache.org/

[ HTTPServer ]
HTTP server header string. This plugin also attempts to
identify the operating system from the server header.

OS : Ubuntu Linux
String : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
PHP is a widely-used general-purpose scripting language
that is especially suited for Web development and can be
embedded into HTML. This plugin identifies PHP errors,
modules and versions and extracts the local file path and
username if present.

Version : 5.2.4-2ubuntu5.10
Google Dorks: (3)
Website : http://www.php.net/

[ WebDAV ]
Web-based Distributed Authoring and Versioning (WebDAV) is
a set of methods based on the Hypertext Transfer Protocol
(HTTP) that facilitates collaboration between users in
```

Figure 11. WhatWeb basic scan and verbose (-v) scan.

```
Session Actions Edit View Help
HTTP Headers:
HTTP/1.1 200 OK
Date: Wed, 02 Sep 2026 05:23:38 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html

[kali@lelahannia:~]$
$ whatweb -a 1 http://192.168.56.104
http://192.168.56.104 [200 OK] Apache[2.2.8], Country[RESERVED][22], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.104], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2u
buntu5.10]

[kali@lelahannia:~]$
$ whatweb -a 3 -v http://192.168.56.104
WhatWeb report for http://192.168.56.104
Status : 200 OK
Title : Metasploitable2 - Linux
IP : 192.168.56.104
Country : RESERVED, 22
Summary : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
The Apache HTTP Server Project is an effort to develop and
maintain an open-source HTTP server for modern operating
systems including UNIX and Windows NT. The goal of this
project is to provide a secure, efficient and extensible
server that provides HTTP services in sync with the current
HTTP standards.

Version : 2.2.8 (from HTTP Server Header)
Google Dorks: (3)
Website : http://httpd.apache.org/

[ HTTPServer ]
HTTP server header string. This plugin also attempts to
identify the operating system from the server header.

OS : Ubuntu Linux
String : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
PHP is a widely-used general-purpose scripting language
that is especially suited for Web development and can be
```

Figure 12. WhatWeb aggression level 1 (compact summary) vs level 3 verbose (full plugin detail).

```
Session Actions Edit View Help
(kali@lelahnaomie)-[~]
$ whatweb -a 4 -v http://192.168.56.104

WhatWeb report for http://192.168.56.104
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.56.104
Country     : RESERVED, ZZ

Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], Matomo, PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

Version      : 2.2.8 (from HTTP Server Header)
Google Dorks: (3)
Website      : http://httpd.apache.org/

[ HTTPServer ]
HTTP server header string. This plugin also attempts to identify the operating system from the server header.

OS           : Ubuntu Linux
String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ Matomo ]
Matomo is the leading open alternative to Google Analytics that gives you full control over your data. Matomo lets you easily collect data from websites, apps & the IoT and visualise this data and extract insights. Privacy is built-in. Matomo was formerly known as Piwik, and is developed in PHP.

Aggressive function available (check plugin file or details).
Google Dorks: (1)
Website      : https://matomo.org

[ PHP ]
PHP is a widely-used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML. This plugin identifies PHP errors, modules and versions and extracts the local file path and username if present.
```

Figure 13. WhatWeb aggression level 4 - note the additional "Matomo" plugin detection not present at lower levels.

```
Session Actions Edit View Help
(kali@lelahnaomie)-[~]
$ whatweb --follow-redirect=always http://192.168.56.104
http://192.168.56.104 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.104], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

(kali@lelahnaomie)-[~]
$ whatweb -v http://192.168.56.104 > whatweb-results.txt
cat whatweb-results.txt
WhatWeb report for http://192.168.56.104
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.56.104
Country     : RESERVED, ZZ

Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

Version      : 2.2.8 (from HTTP Server Header)
Google Dorks: (3)
Website      : http://httpd.apache.org/

[ HTTPServer ]
HTTP server header string. This plugin also attempts to identify the operating system from the server header.

OS           : Ubuntu Linux
String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
PHP is a widely-used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML. This plugin identifies PHP errors, modules and versions and extracts the local file path and username if present.

Version      : 5.2.4-2ubuntu5.10
Google Dorks: (3)
Website      : http://www.php.net/
```

Figure 14. Redirect-follow check (no redirect occurred) and the saved whatweb-results.txt evidence file.

3. Final Service Inventory

Consolidated from the full-port scan (-p- -sV) and the aggressive scan (-A). 28 open TCP ports and 4 open UDP ports were identified in total.

Port	Proto	State	Service	Version / Evidence
21	tcp	open	ftp	vsftpd 2.3.4 (anonymous login allowed)
22	tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23	tcp	open	telnet	Linux telnetd
25	tcp	open	smtp	Postfix smtpd
53	tcp	open	domain	ISC BIND 9.4.2
80	tcp	open	http	Apache httpd 2.2.8 (Ubuntu) DAV/2
111	tcp	open	rpcbind	2 (RPC #100000)
139	tcp	open	netbios-ssn	Samba smbd 3.X-4.X (workgroup: WORKGROUP)
445	tcp	open	netbios-ssn	Samba smbd 3.0.20-Debian
512	tcp	open	exec	netkit-rsh rexecd
513	tcp	open	login	OpenBSD/Solaris rlogind
514	tcp	open	shell	Netkit rshd
1099	tcp	open	java-rmi	GNU Classpath grmiregistry
1524	tcp	open	bindshell	Metasploitable root shell
2049	tcp	open	nfs	2-4 (RPC #100003)
2121	tcp	open	ftp	ProFTPD 1.3.1
3306	tcp	open	mysql	MySQL 5.0.51a-3ubuntu5
3632	tcp	open	distccd	distccd v1 (GNU 4.2.4, Ubuntu)
5432	tcp	open	postgresql	PostgreSQL DB 8.3.0-8.3.7
5900	tcp	open	vnc	VNC protocol 3.3
6000	tcp	open	X11	(access denied)
6667	tcp	open	irc	UnrealIRCd
8009	tcp	open	ajp13	Apache Jserv Protocol v1.3
8180	tcp	open	http	Apache Tomcat/Coyote JSP engine 1.1
8787	tcp	open	drb	Ruby DRb RMI (Ruby 1.8)
43159	tcp	open	java-rmi	GNU Classpath grmiregistry
45393	tcp	open	mountd	1-3 (RPC #100005)
51851	tcp	open	nlockmgr	1-4 (RPC #100021)
52210	tcp	open	status	1 (RPC #100024)
53	udp	open	domain	ISC BIND 9.4.2
111	udp	open	rpcbind	-
137	udp	open	netbios-ns	-
2049	udp	open	nfs	-

OS Fingerprint

Linux kernel 2.6.9-2.6.33, general-purpose device, CPE cpe:/o:linux:linux_kernel:2.6.
Confirmed via smb-os-discovery as Unix/Samba 3.0.20-Debian, hostname "metasploitable",
domain "localdomain".

4. Student Questions

1. What is reconnaissance?

The process of gathering information about a target system or network — live hosts, open ports, running services, versions, and operating system — before any further testing, without exploiting anything. It builds a picture of the attack surface so later testing is targeted rather than blind.

2. What is the difference between host discovery and port scanning?

Host discovery (`nmap -sn`) only determines which IP addresses on a network are alive and reachable; it does not touch any ports. Port scanning then probes a specific live host to find which TCP/UDP ports are open, closed, or filtered on that host.

3. Explain open, closed and filtered.

Open means a service actively accepted the probe (e.g. port 21 replied with a banner). Closed means the host responded but no process is listening there (a TCP RST was received). Filtered means Nmap got no response or an unreachable error, meaning a firewall or ACL is likely blocking the probe, so Nmap cannot tell whether the port is really open or closed.

4. What does -sV do?

It sends targeted probes to each open port and compares the responses against Nmap's service-probe database to identify the actual application and version running (e.g. `vsftpd 2.3.4`), rather than only guessing a service name from the port number.

5. What does -O do?

It performs operating-system fingerprinting by analysing subtle differences in how the target's TCP/IP stack responds to a series of crafted packets (TTL, window size, option ordering, etc.) and matches the pattern against Nmap's OS database. On our target it returned `Linux 2.6.9-2.6.33`, a general-purpose device.

6. Explain what -A combines and why it is noisier.

-A combines OS detection (-O), version detection (-sV), default NSE scripts (-sC), and a traceroute, all in a single scan. It is noisier because it sends far more probes and scripts against every open port (banner grabs, script checks, traceroute packets) than a single-purpose scan, making it easier for an IDS/IPS to flag as an active reconnaissance sweep.

7. What does -p- mean?

It tells Nmap to scan all 65535 TCP ports (1-65535) instead of only the default top 1000, so unusual high-numbered ports are not missed. On our target this revealed several extra ports (e.g. 43159, 45393, 51851, 52210) that the default scan did not show.

8. Why can UDP scans take longer?

UDP has no handshake. A probe to a closed UDP port often gets a fast ICMP port-unreachable reply, but open or filtered ports frequently send no reply at all, so Nmap must wait out a timeout and retransmit before it can decide. In our own scan, the full -sU scan took 1017 seconds (about 17 minutes) versus a few seconds for the equivalent TCP scan.

9. What does -sC do?

It runs Nmap's default set of NSE (Nmap Scripting Engine) scripts — a safe, non-intrusive set chosen by the Nmap developers — against detected services to pull extra information such as banners, anonymous-login checks, and basic service metadata.

10. What information did the HTTP title/headers scripts reveal?

http-title returned "Metasploitable2 - Linux", confirming the web application's identity from its page title alone. http-headers returned the raw response headers: Server: Apache/2.2.8 (Ubuntu) DAV/2, X-Powered-By: PHP/5.2.4-2ubuntu5.10, Content-Type: text/html, confirming the exact web server, Linux distribution, and PHP version without needing a separate -sV scan.

11. What did SMB/SSH/FTP enumeration add?

smb-protocols showed the host still supports SMBv1 ("NT LM 0.12 (SMBv1) [dangerous, but default]"), a legacy protocol with known exploits. ssh-hostkey listed the actual DSA and RSA host key fingerprints in use, confirming SSH is genuinely present and revealing that older DSA key support is itself a weakness indicator. FTP banner enumeration confirmed "220 (vsFTPd 2.3.4)" matching the -sV result exactly, and the wider -A scan additionally revealed that anonymous FTP login is allowed (FTP code 230) — a concrete, actionable finding beyond just a version string.

12. What is the difference between Nmap service detection and WhatWeb fingerprinting?

Nmap -sV works at the network/service level: it probes the port itself and reads the raw response or banner to identify what is listening (e.g. Apache httpd 2.2.8). WhatWeb works at the application/content level: it fetches the actual web page and inspects the HTML, headers, cookies, and embedded technologies to identify frameworks, CMSs, and libraries running behind that web server (e.g. PHP 5.2.4, WebDAV 2, and a Matomo detection at higher aggression). The two are complementary: Nmap identifies what is serving the page, WhatWeb identifies what is built on top of it.

13. How did WhatWeb levels 1, 3 and 4 differ?

Level 1 (default aggression) returned a compact single-line summary: Apache, HTTPServer, WebDAV, PHP, X-Powered-By. Level 3 added the full multi-paragraph verbose report explaining each detected plugin in detail. Level 4 (maximum aggression) went further still and additionally flagged "Matomo" as a possible plugin, a result not present at the lower levels, showing that higher aggression levels attempt more speculative detection techniques that can surface extra findings at the cost of sending more requests to the target and occasionally producing false positives.

14. Why must aggressive scanning/fingerprinting remain inside the authorised lab?

Aggressive scans (-A, full port sweeps, WhatWeb level 4) send a high volume of probes, some of which resemble attack traffic (rapid connection attempts, crafted OS-fingerprint packets, repeated banner grabs). Outside an authorised, isolated lab this traffic could be indistinguishable from malicious activity, disrupt production services, breach unauthorised-computer-access laws, or expose real systems to unnecessary risk. Restricting it to an authorised, isolated target such as this Metasploitable2 VM on a host-only network ensures the findings are safe to generate and legal to produce.

